Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and Labor Markets — qualifier level (Listas 3–4)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — Consumption, present bias and interest rates

(Kurlat chs. 6, 9; Romer ch. 8 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household lives $T+1$ periods ($t = 0, \ldots, T$; $T \geq 1$) with $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$ ($\ln c$ if $\sigma = 1$), discount factor $0 < \beta < 1$ and a constant real rate $r > 0$; its total wealth (financial plus human, in present value at $t = 0$) is W. The share c_0/W is:

A. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = \beta^{\sigma}(1+r)^{\sigma-1}$

B. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = \beta^{1/\sigma}(1+r)^{1/\sigma-1}$

C. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = \beta$

D. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = [\beta(1+r)]^{\sigma}$

Answer: B

Step — path and budget. The Euler equation (6.3.9), $c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma}$, gives $c_t = [\beta(1+r)]^{t/\sigma}c_0$. Leaving assets after T is wasteful, so (6.3.6) binds: $\sum_{t=0}^{T} c_t(1+r)^{-t} = c_0\sum_{t=0}^{T}\gamma^t = W$ with $\gamma \equiv \beta^{1/\sigma}(1+r)^{1/\sigma-1}$. Summing the geometric series, $c_0/W = (1-\gamma)/(1-\gamma^{T+1})$ ($1/(T+1)$ as $\gamma \to 1$). Mistakes: A takes σ for the intertemporal elasticity $1/\sigma$; C is the log case, right only if $\sigma = 1$ or $\beta(1+r) = 1$; D makes two mistakes: it takes σ for $1/\sigma$ and does not discount consumption growth in the budget.

Ref: Kurlat (2020), ch. 6, §6.3, eqs. (6.3.6) and (6.3.9).

b) (3 points) *Consumption × general equilibrium.* A two-period endowment economy with a representative household: endowments $y_1 > 0$ and $y_2 = (1+g)y_1$, $g > 0$; preferences $u(c_1) + \beta u(c_2)$, CRRA with $\sigma > 0$, $0 < \beta < 1$; a bond in zero net supply pays r. The equilibrium real interest rate is:

A. $r = (1+g)/\beta - 1$

B. $r = 1/\beta - 1$

C. $r = (1+g)^{1/\sigma} - 1$

D. $r = (1+g)^{\sigma}/\beta - 1$

Answer: D

Step — clearing and price. A representative household holds none of a bond in zero net supply, so $c_t = y_t$. The Euler equation (9.1.8), $c_1^{-\sigma} = \beta(1+r)c_2^{-\sigma}$, then gives $1 + r = (y_2/y_1)^{\sigma}/\beta = (1+g)^{\sigma}/\beta$: the rate makes the household content with its growing endowment. Mistakes: A is the log case (right only if $\sigma = 1$); B drops growth (right only if $g = 0$); C makes two mistakes: it takes σ for the elasticity $1/\sigma$ and drops the discount factor (as if $\beta = 1$).

Ref: Kurlat (2020), ch. 9, §9.1, eq. (9.1.8), with endowments in place of production.

c) (2 points, T or F) *Consumption × credit markets.* Two periods, endowments (y_1, y_2), utility $u(c_1) + \beta u(c_2)$ with u strictly increasing and strictly concave; the household saves at r_s and borrows at $r_b > r_s$. At the current rates it neither borrows nor lends, and the first unit saved would strictly lower its utility at the margin. Then any rise in r_s, however small, makes it save.

Answer: F

Step — kink. Let $m \equiv u'(y_1)/[\beta u'(y_2)]$. Saving $a \geq 0$ yields $f(a) = u(y_1 - a) + \beta u(y_2 + (1 + r_s)a)$, strictly concave, with $f'(0) = \beta u'(y_2)(1 + r_s - m)$; the premise is $f'(0) < 0$, i.e. $m > 1 + r_s$. At a new rate r_s' with $1 + r_s' < m$, still $f'(0) < 0$, so $f'(a) < 0$ for all $a \geq 0$, and borrowing does not involve r_s: the household stays put. Only rises beyond $m - (1 + r_s)$ make it save.

Ref: Kurlat (2020), ch. 6, §6.2 and Exercise 6.6 (a tax on savings kinks the budget set).

d) (2 points, T or F) *Consumption × fiscal policy.* Two periods, utility $u(c_1) + \beta u(c_2)$ with u strictly increasing and strictly concave, income y_t, lump-sum taxes τ_t; the household's saving faces the limit $a \geq -b$, $b \geq 0$, which binds. A tax cut $\Delta > 0$ today is financed by a tax increase $(1 + r)\Delta$ next period. Then current consumption rises by exactly Δ, for every $\Delta > 0$.
Answer: F

Step — threshold. After the cut $c_2 = (1 + r)(W - c_1)$ with $W \equiv y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}$: the budget line does not move, so the strictly concave $u(c_1) + \beta u((1 + r)(W - c_1))$ has a maximizer c_1^u free of Δ. Only the limit moves, $c_1 \leq y_1 - \tau_1 + b + \Delta$, so $c_1 = \min\{c_1^u,\ y_1 - \tau_1 + b + \Delta\}$. The limit binds before the cut: $\bar{\Delta} \equiv c_1^u - (y_1 - \tau_1 + b) > 0$, and $\Delta c_1 = \min\{\Delta, \bar{\Delta}\}$: **False** for every $\Delta > \bar{\Delta}$, whatever u and β.

Step — reading. Up to $\bar{\Delta}$ the cut is the loan the household could not get, spent one for one; a larger cut makes the limit slack, and the household saves the rest to pay the later tax (Ricardian equivalence at the margin).

Ref: Kurlat (2020), ch. 6, §6.2 and Exercise 6.5(e).

Question 2. (15 points) Discursive. *Consumption × behavioral economics × general equilibrium.* Three periods $t = 0, 1, 2$; in (a)–(c) the real interest rate is zero and the household has wealth $W > 0$ at date 0. The self at date t maximizes $u(c_t) + \beta \sum_{s>t} u(c_s)$ with $u(c) = c^{1-\sigma}/(1 - \sigma)$, $\sigma > 0$ ($\ln c$ if $\sigma = 1$), and $0 < \beta < 1$.

a) (3 points) Find the plan chosen with commitment (self 0 chooses all three consumptions), by a naive self 0 (who believes future selves will follow its plan) and by a sophisticated self 0 (who anticipates self 1's choice). In what sense are these preferences time-inconsistent?

Step — self 1. With $W_1 = W - c_0$, self 1 solves $\max u(c_1) + \beta u(c_2)$ s.t. $c_1 + c_2 = W_1$. The FOC $c_1^{-\sigma} = \beta c_2^{-\sigma}$ gives $c_2 = \omega c_1$ with $\omega \equiv \beta^{1/\sigma} \in (0, 1)$, so $c_1 = kW_1$ and $c_2 = (1 - k)W_1$, $k \equiv 1/(1 + \omega) \in (\frac{1}{2}, 1)$.

Step — commitment and naivete. Under $c_0 + c_1 + c_2 = W$, self 0's FOCs $u'(c_1) = u'(c_2)$, $u'(c_0) = \beta u'(c_1)$ give $c_1 = c_2 = \omega c_0$: $(c_0^C, c_1^C, c_2^C) = \frac{W}{1+2\omega}(1, \omega, \omega)$. The naive self 0 expects this plan, so $c_0^N = c_0^C$, and self 1 splits $W_1 = \frac{2\omega W}{1+2\omega}$: $(c_1^N, c_2^N) = \frac{2\omega W}{(1+2\omega)(1+\omega)}(1, \omega)$.

Step — sophistication. By homogeneity, $u(sW_1) = s^{1-\sigma}u(W_1)$, so self 1's rule leaves self 0 the continuation $\beta A(k)u(W_1)$, with $A(s) \equiv s^{1-\sigma} + (1 - s)^{1-\sigma}$ (under log, $A \equiv 2$ up to a constant). The FOC of the strictly concave $\max_{c_0} u(c_0) + \beta A(k)u(W - c_0)$ is $c_0^{-\sigma} = \beta A(k)(W - c_0)^{-\sigma}$; with $k = \frac{1}{1+\omega}$, $1 - k = \frac{\omega}{1+\omega}$ and $\omega^\sigma = \beta$,

$$c_0^S = \frac{W}{1 + \Lambda}, \qquad \Lambda \equiv [\beta A(k)]^{1/\sigma} = (1 + \omega)^{1-1/\sigma}(\beta + \omega)^{1/\sigma}, \qquad (c_1^S, c_2^S) = \frac{\Lambda W}{(1 + \Lambda)(1 + \omega)}(1, \omega).$$

Under log, $\Lambda = 2\beta$: the sophisticated plan is the naive one.

Economic intuition. $c_1^N/c_1^C = \frac{2}{1+\omega} > 1 > \frac{2\omega}{1+\omega} = c_2^N/c_2^C$: with nothing new learned, self 1 overturns the plan, since the weight of c_2 relative to c_1 is 1 seen from date 0 and β seen from date 1.

Ref: Romer (2012), ch. 8, §8.6 (time-inconsistent preferences) and Problem 8.15; Kurlat (2020), ch. 6, §6.4.

Grading rubric.

- Self 1's rule 0.5; commitment 0.5; naive 0.5; sophisticated (continuation, FOC, plan) 1; inconsistency 0.5.
- Log case solved in place of general σ: max. half. Plans quoted (e.g. from Romer 8.15), not derived: max. a third.

b) (3 points) Determine when the sophisticated self 0 consumes more, less or the same as the naive one, and explain the mechanism.

Step — reduction. From (a), $c_0^S \lesseqgtr c_0^N \iff \Lambda \lesseqgtr 2\omega \iff \beta A(k) \lesseqgtr (2\omega)^\sigma = \beta 2^\sigma \iff A(k) \lesseqgtr A(\frac{1}{2})$, as $A(\frac{1}{2}) = 2 \cdot 2^{\sigma-1} = 2^\sigma$.

Step — Jensen. $A(k) = \varphi(k) + \varphi(1-k)$ with $\varphi(z) = z^{1-\sigma}$, $\varphi''(z) = \sigma(\sigma-1)z^{-\sigma-1}$, and $k \neq \frac{1}{2}$ (as $\omega < 1$) with midpoint $\frac{1}{2}$. Strict Jensen: $\sigma > 1$ (φ convex) $\Rightarrow A(k) > A(\frac{1}{2}) \Rightarrow c_0^S < c_0^N$; $\sigma < 1$ (φ concave) $\Rightarrow c_0^S > c_0^N$; $\sigma = 1$ ($A \equiv 2$) $\Rightarrow c_0^S = c_0^N$.

Economic intuition. Let $R \equiv [A(k)/A(\frac{1}{2})]^{1/(1-\sigma)}$, so $A(k) = A(\frac{1}{2})R^{1-\sigma}$: the sophisticate's FOC is a naive self's FOC with a gross return $R < 1$ on saving (in both cases above). A lower return raises saving iff the income effect dominates ($\sigma > 1$) and lowers it iff the substitution effect does ($\sigma < 1$); under log they cancel and the split drops out.

Ref: Romer (2012), Problem 8.15(b)(ii) and ch. 8, §8.4; Kurlat (2020), Exercise 6.1(e).

Grading rubric.

- Reduction to $A(k)$ versus $A(\frac{1}{2})$ 1 pt; Jensen, three cases 1 pt; mechanism (effective return) 1 pt.
- Ordering asserted without the convexity argument: 0 for that line. Only the log case treated: max. half.

c) (3 points) At date 0 self 0 can buy an illiquid account that locks in the (c_1, c_2) it chooses. What share x of W would a sophisticated self 0 pay for it at most? Show whether x depends on W, bound it, give it under log utility and find how it behaves as $\beta \to 1$. What would a naive self 0 pay?

Step — values. For weights $w_j > 0$, $\max \sum_j w_j u(c_j)$ s.t. $\sum_j c_j = W$ has FOCs $w_j c_j^{-\sigma} = \lambda$, so $c_j = W w_j^{1/\sigma}/S$, $S \equiv \sum_j w_j^{1/\sigma}$, and value $\frac{W^{1-\sigma}}{1-\sigma} S^\sigma$. Commitment has weights $(1, \beta, \beta)$, $S = 1 + 2\omega$; the sophisticated reduced problem of (a) has $(1, \beta A(k))$, $S = 1 + \Lambda$.

Step — price of commitment. x solves $V_C((1-x)W) = V_S(W)$, i.e. $(1-x)^{1-\sigma}(1+2\omega)^\sigma = (1+\Lambda)^\sigma$:

$$x = 1 - \Big(\frac{1+\Lambda}{1+2\omega}\Big)^{\sigma/(1-\sigma)},$$

independent of W. As the commitment plan is self 0's unique optimum and $c_1^S \neq c_2^S$, $V_S(W) < V_C(W)$; V_C is continuous and strictly increasing with $V_C(0^+) < V_S(W)$, so the root is unique and $0 < x < 1$.

Step — log. For $\sigma = 1$, $V_j(W) = (1+2\beta)\ln W + v_j$ and, as $c_0^S = c_0^C$, $v_C - v_S = \beta \ln \frac{1/4}{k(1-k)} = \beta \ln \frac{(1+\beta)^2}{4\beta}$, so $x = 1 - [\frac{4\beta}{(1+\beta)^2}]^{\beta/(1+2\beta)}$, the limit of the general x as $\sigma \to 1$ (write $u = \frac{c^{1-\sigma}-1}{1-\sigma} \to \ln c$, which shifts both sides of the indifference condition equally).

Step — $\beta \to 1$. Write $\Lambda = 2\omega\mu$, $\mu \equiv [A(k)/A(\frac{1}{2})]^{1/\sigma}$, $\delta \equiv k - \frac{1}{2} = \frac{1-\omega}{2(1+\omega)}$. A is symmetric about $\frac{1}{2}$, so $\frac{A(1/2+\delta)}{A(1/2)} = 1 + 2\sigma(\sigma-1)\delta^2 + O(\delta^4)$, $\mu = 1 + 2(\sigma-1)\delta^2 + O(\delta^4)$ and $\ln(1-x) = \frac{\sigma}{1-\sigma}\ln[1 + \frac{2\omega(\mu-1)}{1+2\omega}] = -\frac{4\sigma\omega}{1+2\omega}\delta^2 + O(\delta^4)$. With $\omega = 1 - \frac{1-\beta}{\sigma} + O((1-\beta)^2)$ and $\delta = \frac{1-\beta}{4\sigma} + O((1-\beta)^2)$,

$$x = \frac{(1-\beta)^2}{12\sigma} + O\big((1-\beta)^3\big),$$

also for $\sigma = 1$ (expand the log formula).

Step — naive. A naive self 0 expects to follow its plan anyway, so it would pay nothing. Its actual loss is larger: $V_N(W)$ evaluates at c_0^N the strictly concave objective that c_0^S maximizes, so $V_N \leq V_S$, with equality iff $\sigma = 1$.

Economic intuition. The account buys back the even split of the future that self 0 wants. Self 1's tilt costs self 0 utility only to second order, so a small bias makes commitment nearly worthless; and only a self that foresees the tilt pays.

Ref: Romer (2012), Problem 8.15; Kurlat (2020), ch. 6, §6.4 (401(k) defaults).

Grading rubric.

- Values 1 pt; x, independence of W, $0 < x < 1$ 1 pt; log case, order in $1 - \beta$, naive answer 1 pt.

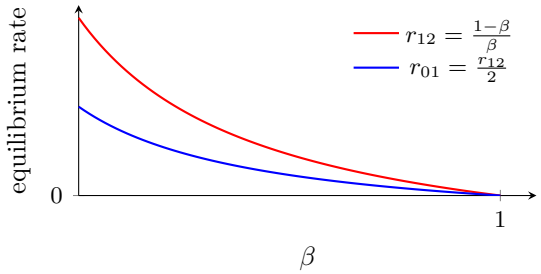
- Log formula used for general σ: max. half. Order or bound claimed without derivation: 0 for that line.

d) (3 points) Now each household receives $y > 0$ at each date and has no initial wealth; households are identical and sophisticated, and borrow and lend at r_{01} (from date 0 to 1) and r_{12} (from 1 to 2). Find the equilibrium interest rates, show whether they depend on σ, and draw them against β.

Step — clearing and r_{12}. Identical households cannot lend to each other, so nobody saves: $c_0 = c_1 = c_2 = y$. Self 1 maximizes $u(c_1) + \beta u(c_2)$ s.t. $c_1 + \frac{c_2}{1+r_{12}} = W_1$; its Euler equation $u'(c_1) = \beta(1+r_{12})u'(c_2)$ at $c_1 = c_2 = y$ gives $1 + r_{12} = 1/\beta$.

Step — r_{01}. At that price self 1 sets $c_2 = c_1$ for every W_1, so the budget $c_1 + \beta c_2 = W_1$ gives $c_1 = c_2 = \frac{W_1}{1+\beta}$, whatever σ. Self 0 maximizes $u(c_0) + 2\beta u(\frac{W_1}{1+\beta})$ with $W_1 = (1+r_{01})(y-c_0) + (1+\beta)y$; its FOC $u'(c_0) = (1+r_{01})\frac{2\beta}{1+\beta}u'(\frac{W_1}{1+\beta})$ at $c_0 = y$, $W_1 = (1+\beta)y$, gives $1 + r_{01} = \frac{1+\beta}{2\beta}$. Neither rate depends on σ (all marginal utilities are $u'(y)$; the propensities are $\frac{1}{1+\beta}$):

$$r_{01} = \frac{1-\beta}{2\beta}, \qquad r_{12} = \frac{1-\beta}{\beta} = 2r_{01}.$$



Economic intuition. Rates rise until each self is content with its endowment. Self 1 discounts date 2 by β, so $r_{12} = 1/\beta - 1$. Self 0 discounts date 1 by β too, but values dates 1 and 2 equally, and a unit saved to date 1 is spread over both at the high rate r_{12}: it needs only half the premium.

Ref: Kurlat (2020), ch. 9, §9.1, eq. (9.1.8) (an endowment economy); Romer (2012), ch. 8, §8.6.

Grading rubric.

- Clearing and r_{12} 1 pt; propensities and r_{01} 1 pt. Rates from an Euler equation without clearing: max. half.
- Graph, scored separately (1 pt): labelled axes; two decreasing curves vanishing at $\beta = 1$; $r_{12} = 2r_{01}$ above.

e) (3 points) *Parameter analysis.* Sign the effects of β, of σ and of naivete (all households naive instead of sophisticated, also in (d)) on x, r_{01} and r_{12}; prove each sign and collect the signs in a table, give the limits of x at the ends of the admissible ranges, and read the results.

Result	$\beta \uparrow$	$\sigma \uparrow$	naive instead of sophisticated
x (value of commitment)	$-$ near $\beta = 1^\dagger$	$-$ near $\beta = 1^\dagger$	perceived value 0
r_{01}	$-$	0	sign of $\sigma - 1$
r_{12}	$-$	0	0

†Not monotone: $x \to 0$ at both ends of each range.

Step — x at the ends. For $\sigma \neq 1$, $1 - x = B^\eta$ with $B \equiv \frac{1+\Lambda}{1+2\omega}$, $\eta \equiv \frac{\sigma}{1-\sigma}$. $\beta \to 1$: $\omega \to 1$, $\Lambda \to 2$. $\beta \to 0$: $\omega \to 0$, $\Lambda \to 0$ ($\beta + \omega \to 0$). $\sigma \to \infty$: $\omega \to 1$, $\Lambda \to 2$, $\eta \to -1$. $\sigma \to 0$: $\omega \to 0$, $\Lambda = \omega(1+\omega)^{1-1/\sigma}(1 + \frac{\omega}{\beta})^{1/\sigma} \leq \omega e^{\omega/(\beta\sigma)} \to 0$, $\eta \to 0$. Each time $B \to 1$ with η bounded, so $x \to 0$; under log too ($\beta \ln \beta \to 0$ as $\beta \to 0$).

Step — x: signs. x is continuous, positive inside and vanishes at both ends of each range, so it has an interior maximum in β and in σ: neither effect has a global sign. Near $\beta = 1$, differentiating the smooth expansion of (c): $\partial_\beta x = -\frac{1-\beta}{6\sigma} + O((1-\beta)^2) < 0$ and $\partial_\sigma x = -\frac{(1-\beta)^2}{12\sigma^2} + O((1-\beta)^3) < 0$.

Step — rates. $\partial_\beta r_{01} = -\frac{1}{2\beta^2} < 0$, $\partial_\beta r_{12} = -\frac{1}{\beta^2} < 0$, $\partial_\sigma r_{01} = \partial_\sigma r_{12} = 0$; both vanish as $\beta \to 1$ and diverge as $\beta \to 0$.

Step — naive economy. Self 1 is unchanged, so is r_{12}. A naive self 0 believes self 1 will maximize $u(c_1) + u(c_2)$ s.t. $c_1 + \beta c_2 = W_1$: planned $c_1^p = \frac{W_1}{1+\beta^p}$, $p \equiv 1 - \frac{1}{\sigma}$, and by the envelope

theorem W_1 is worth $\beta u'(c_1^p)$ at the margin. Its Euler equation at $c_0 = y$, $W_1 = (1+\beta)y$ gives $1 + r_{01}^N = \frac{1}{\beta}(\frac{1+\beta}{1+\beta^p})^\sigma$, which depends on σ. Against $1 + r_{01}^S = \frac{1+\beta}{2\beta}$, taking $1/\sigma$ powers: $r_{01}^N \gtrless r_{01}^S \iff (\frac{1+\beta}{2})^p \gtrless \frac{1+\beta^p}{2}$. As z^p is strictly concave for $p \in (0,1)$ ($\sigma > 1$) and strictly convex for $p < 0$ ($\sigma < 1$), $\text{sign}(r_{01}^N - r_{01}^S) = \text{sign}(\sigma - 1)$. Knife-edge $\sigma = 1$: naivete changes nothing but the perceived x.

Economic intuition. Commitment is worth most at intermediate present bias: it vanishes when the selves agree ($\beta \to 1$, a Harberger triangle), when self 0 ignores the future ($\beta \to 0$), when self 1 splits evenly ($\sigma \to \infty$) and when every self consumes at once ($\sigma \to 0$). Sophisticates buy commitment; naifs, who lose more, do not: the case for defaults and forced saving (401(k) defaults, Brazil's FGTS). Present bias raises real rates, $r_{12} = 2r_{01}$, free of σ under sophistication; naivete moves r_{01} only if $\sigma \neq 1$.

Ref: Kurlat (2020), ch. 6, §6.4 and ch. 9, §9.1; Romer (2012), ch. 8, §8.4 and §8.6.

Grading rubric.

- Signs proved, by row, the naive comparison included: 1.5 pts. Table: 0.5 pt. Limits of x: 0.5 pt. Reading: 0.5 pt.
- Signs without proofs or reading: max. half. A limit or sign claimed without proof: 0 for that line.

Block 2 — Labor supply, taxes, transfers and search

(Kurlat ch. 7; Romer ch. 10 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Labor × consumption smoothing.* A household lives $N \geq 2$ periods with per-period utility $\ln c_t - \psi\, h_t^{1+1/\varepsilon}/(1+1/\varepsilon)$, $\psi, \varepsilon > 0$; it has no initial wealth, $\beta = 1$ and $r = 0$, and the wage is the same in every period. The wage rises by the same small proportion in k of the periods, $1 \leq k < N$. To first order, the elasticity of hours in a period with the raise with respect to its wage is:

A. ε

B. $\varepsilon/(1+\varepsilon)$

C. $\varepsilon\,(N-k)/N$

D. $\varepsilon\,(N-k)/(N+k\varepsilon)$

Answer: C

Step — FOCs. With $\beta(1+r) = 1$ the Euler equation gives $c_t = c$ in every period, and the hours condition $\psi h_t^{1/\varepsilon} = w_t/c$ gives $d\ln h_t = \varepsilon\,(d\ln w_t - d\ln c)$.

Step — budget. $Nc = \sum_t w_t h_t$. At equal wages each period earns the share $1/N$, so $d\ln c = \frac{1}{N}\sum_t (d\ln w_t + d\ln h_t)$. Substituting the FOC, $(1+\varepsilon)\,d\ln c = (1+\varepsilon)\frac{k}{N}\,d\ln w$, so $d\ln c = \frac{k}{N}\,d\ln w$ (the wealth effect is the share of earnings the raise touches): hours rise by $\varepsilon\frac{N-k}{N}\,d\ln w$ in a raised period and fall by $\varepsilon\frac{k}{N}\,d\ln w$ in the others.

Step — distractors. A holds the marginal utility of wealth fixed (pure Frisch response); B is the static compensated elasticity of these preferences, with no intertemporal margin; D keeps only the raised periods' hours in the budget. A permanent raise ($k = N$) gives 0: log utility then cancels income and substitution effects.

Ref: Kurlat (2020), §7.4, and Exercise 7.5(j)–(k) for the Frisch elasticity.

b) (3 points) *Search × wage setting.* Steady-state search model with matches $M = A_m U^\xi V^{1-\xi}$, $\xi \in (0,1)$, tightness $\theta = V/U$, vacancy-filling rate $q(\theta) = M/V$ and free entry $(y-w)\,q(\theta) = (r+s)\kappa$, with $r, s, \kappa > 0$. The wage is rigid at $w \in (0, y)$ while productivity y moves. The elasticity of θ with respect to y is:

A. $\frac{1}{\xi}\,\frac{y}{y-w}$

B. $\frac{1}{1-\xi}\,\frac{y}{y-w}$

C. $\frac{1}{1-\xi}$

D. $\frac{1-\xi}{\xi}\,\frac{y}{y-w}$

Answer: A

Step — matching. $q(\theta) = A_m U^{\xi} V^{-\xi} = A_m \theta^{-\xi}$ and $f = \theta q = A_m \theta^{1-\xi}$, so free entry reads $\ln(y-w) - \xi \ln\theta = \ln\frac{(r+s)\kappa}{A_m}$.

Step — log-differentiate. With w fixed, $\frac{y}{y-w}\, d\ln y = \xi\, d\ln\theta$, so $\frac{d\ln\theta}{d\ln y} = \frac{1}{\xi}\,\frac{y}{y-w}$: through the profit $y-w$ a rigid wage amplifies y by $\frac{y}{y-w} > 1$ (one answer to Shimer's puzzle). B uses $1-\xi$, the elasticity of f, for that of q; C makes two mistakes: it uses $1-\xi$ for ξ, as B does, and lets w move in proportion to y, removing the amplification; D is the elasticity of f, $(1-\xi)\, d\ln\theta / d\ln y$.

Ref: Kurlat (2020), §7.5; Romer (2012), §10.6 and problem 10.15.

c) (2 points, T or F) *Search × efficiency.* One-shot matching: U unemployed workers and V vacancies, each costing $\kappa > 0$ to open (free entry), form $M(U,V)$ matches, with M of constant returns, increasing and strictly concave in V. A match produces y, of which the worker gets the share $\phi \in (0,1)$. The equilibrium maximizes net output $yM(U,V) - \kappa V$ if and only if ϕ equals the elasticity of M with respect to U at the equilibrium tightness.

Answer: T

Step — two conditions. With $\theta = V/U$, $m(\theta) = M(1,\theta)$ and $q = m/\theta$, free entry is $(1-\phi)\, y\, q(\theta^e) = \kappa$; the planner's $W(V) = yU\, m(V/U) - \kappa V$ is strictly concave, maximized at the unique root of $W'(V) = y\, m'(\theta) - \kappa$.

Step — wedge. With $\eta_V \equiv \theta m'/m$, $m' = \eta_V q$, and free entry gives $W'(V^e) = y\, q(\theta^e)\,[\eta_V(\theta^e) - (1-\phi)]$: V^e is efficient iff $\eta_V(\theta^e) = 1-\phi$ (too many vacancies if $1-\phi > \eta_V$).

Step — constant returns. Euler's theorem, $UM_U + VM_V = M$, gives $\eta_U = 1 - \eta_V$: the condition is $\phi = \eta_U(\theta^e)$, **True**. With returns of degree $d \neq 1$, $\eta_U + \eta_V = d$ and only the vacancy version holds.

Ref: Romer (2012), §10.7, eqs. (10.71)–(10.75), pp. 497–498, and problem 10.16; Hosios (1990).

d) (2 points, T or F) *Labor × transfers.* With $u = \ln c + \alpha \ln(1-h)$, $\alpha > 0$, a household that receives a lump-sum transfer $T > 0$ (taken as given) works the same hours at every after-tax wage w_n.

Answer: F

Step — supply. $\max_h \ln(w_n h + T) + \alpha\ln(1-h)$ gives $\frac{w_n}{w_n h + T} = \frac{\alpha}{1-h}$, so $h = \frac{1}{1+\alpha} - \frac{\alpha T}{(1+\alpha)w_n}$ (interior iff $w_n > \alpha T$) and $\frac{\partial h}{\partial w_n} = \frac{\alpha T}{(1+\alpha)w_n^2} > 0$ for $T > 0$: **False**.

Step — why. Leisure is $1 - h = \frac{\alpha}{1+\alpha}\,\frac{w_n + T}{w_n}$ (full income over its price): the effects cancel only if full income is proportional to w_n, i.e. $T = 0$. With $T < 0$ (a lump-sum tax) hours fall with w_n.

Ref: Kurlat (2020), §7.2, and Exercise 7.5(b)–(c).

Question 4. (15 points) Discursive. *Labor supply × taxes and transfers × welfare.* Identical households have $u = \ln c + \alpha\ln(1-h)$, $\alpha > 0$, and one unit of time; a linear technology pays a wage of one per hour. Labor income is taxed at $\tau \in [0,1)$ and the revenue is returned as a lump-sum transfer T that each household takes as given.

a) (3 points) Derive individual labor supply given (τ, T). Show what the tax does when $T = 0$, and find equilibrium hours $h(\tau)$ when the budget balances.

Step — household problem. Given (τ, T): $\max_h \ln[(1-\tau)h + T] + \alpha\ln(1-h)$ is strictly concave, so the FOC $\frac{1-\tau}{c} = \frac{\alpha}{1-h}$, with $c = (1-\tau)h + T$, is sufficient.

Step — individual supply. $\alpha[(1-\tau)h + T] = (1-\tau)(1-h)$ gives $h = \frac{1}{1+\alpha} - \frac{\alpha T}{(1+\alpha)(1-\tau)}$, interior iff $T < \frac{1-\tau}{\alpha}$. With $T = 0$, $h = \frac{1}{1+\alpha}$ at every τ: the tax's income and substitution effects cancel.

Step — balanced budget. Impose $T = \tau h$ after the FOC (each household takes T as given): $c = h$, and $\frac{1-\tau}{h} = \frac{\alpha}{1-h}$ gives

$$h(\tau) = \frac{1-\tau}{1+\alpha-\tau}, \qquad h'(\tau) = -\frac{\alpha}{(1+\alpha-\tau)^2} < 0, \qquad h(0) = \frac{1}{1+\alpha}.$$

Imposed before the FOC, it would remove the wedge: $h = \frac{1}{1+\alpha}$.

Economic intuition. The rebate returns the income the tax takes, leaving only the substitution effect.

Ref: Kurlat (2020), §7.2–7.3 and Exercise 7.5(a)–(e).

Grading rubric.

- FOC, sufficiency: 1 pt. Supply and $T = 0$: 1 pt. $h(\tau)$, T imposed after the FOC: 1 pt.
- Caps: T imposed before the FOC, max. 1.5 pts; quoted, not derived, max. 1 pt.

b) (3 points) Find the tax rate τ^* that maximizes revenue $R(\tau)$ (the size of the transfer program) and the maximal revenue $R^* \equiv R(\tau^*)$, both in closed form; prove that revenue is single-peaked, and draw $R(\tau)$ for two values $\alpha < \alpha'$.

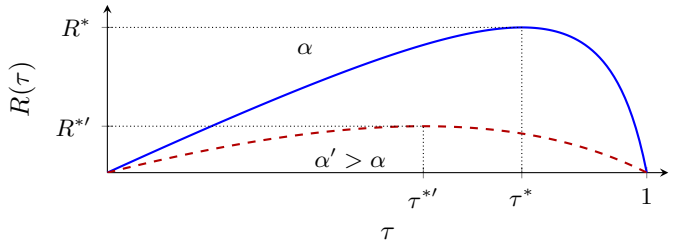
Step — revenue. $R(\tau) = \tau h(\tau) = \frac{\tau(1-\tau)}{1+\alpha-\tau}$, zero at $\tau = 0$ and as $\tau \to 1$, and $R'(\tau) = \frac{N(\tau)}{(1+\alpha-\tau)^2}$ with $N(\tau) = \tau^2 - 2(1+\alpha)\tau + (1+\alpha)$.

Step — single peak. N is a convex parabola with $N(0) = 1+\alpha > 0$ and $N(1) = -\alpha < 0$: one root in $(0,1)$; the other, $(1+\alpha) + \sqrt{\alpha(1+\alpha)}$, exceeds 1. So $R' > 0$ below and $R' < 0$ above

$$\tau^* = (1+\alpha) - \sqrt{\alpha(1+\alpha)} = \sqrt{1+\alpha}\,(\sqrt{1+\alpha} - \sqrt{\alpha}), \qquad R^* = (\sqrt{1+\alpha} - \sqrt{\alpha})^2 = \frac{1}{(\sqrt{1+\alpha} + \sqrt{\alpha})^2},$$

using $1 - \tau^* = \sqrt{\alpha}\,(\sqrt{1+\alpha} - \sqrt{\alpha})$ and $1 + \alpha - \tau^* = \sqrt{\alpha(1+\alpha)}$ in $R^* = \tau^* h(\tau^*)$.

Step — graph. R is hump-shaped, peaking at (τ^*, R^*); $\partial R/\partial\alpha = -\frac{\tau(1-\tau)}{(1+\alpha-\tau)^2} < 0$ puts the α' curve below, peaking left (e).



Economic intuition. Revenue peaks where the base falls as fast, in percent, as the rate rises: $e_{BB}\,\tau = 1 - \tau$ (see (d)).

Ref: Kurlat (2020), §7.3 and Exercise 7.5(e).

Grading rubric.

- R, sign of R': 1 pt. Single peak, τ^*, R^*: 1 pt. Cap: unproved sign of R', 0 for it.
- Graph (scored apart): axes, humps, peaks marked, the α' curve lower and peaking left: 1 pt.

c) (3 points) Derive in closed form the welfare cost of the program in consumption-equivalent terms: the share $x(\tau)$ of consumption at $\tau = 0$ the household would give up to avoid the program. Determine its order at $\tau = 0$ and its monotonicity in τ; find its leading term around $\tau = 0$ and its limit as $\tau \to 1$.

Step — welfare. With $c = h$, $U(\tau) = \ln h(\tau) + \alpha \ln(1 - h(\tau))$, and x solves $\ln[(1-x)\,c(0)] + \alpha \ln(1 - h(0)) = U(\tau)$. With $h = \frac{1-\tau}{1+\alpha-\tau}$ and $1 - h = \frac{\alpha}{1+\alpha-\tau}$:

$$\ln(1-x) = \ln(1-\tau) + (1+\alpha)\ln\frac{1+\alpha}{1+\alpha-\tau}, \qquad x(\tau) = 1 - (1-\tau)\Big(\frac{1+\alpha}{1+\alpha-\tau}\Big)^{1+\alpha}.$$

Step — shape. $\frac{d}{d\tau}\ln(1-x) = -\frac{1}{1-\tau} + \frac{1+\alpha}{1+\alpha-\tau} = -\frac{\alpha\tau}{(1-\tau)(1+\alpha-\tau)}$: zero at $\tau = 0$ (no first-order cost) and negative on $(0,1)$, so x is strictly increasing. As $\tau \to 1$, $1-\tau \to 0$ while the other factor tends to $(\frac{1+\alpha}{\alpha})^{1+\alpha}$: $x \to 1$.

Step — leading term. The derivative is $-\frac{\alpha}{1+\alpha}\tau + O(\tau^2)$; integrating from $\ln(1-x(0)) = 0$, $x(\tau) = \frac{\alpha}{1+\alpha}\frac{\tau^2}{2} + O(\tau^3)$: Harberger's triangle $\frac{1}{2}e_H\tau^2$, with $e_H = \frac{\alpha}{1+\alpha}$ the Hicksian elasticity at $\tau = 0$ (see (d)).

Economic intuition. The program moves nothing between identical households; its whole effect is the wedge on the price of leisure, a pure substitution effect, second order near zero.

Ref: Kurlat (2020), Exercise 7.5(g)–(h).

Grading rubric.

- Closed form: 1 pt. First-order zero, monotonicity, limit, proved: 1 pt. Leading term, e_H: 1 pt.
- Caps: unproved sign or limit, 0 for it; revenue thrown away (one model only), max. half.

d) (3 points) At the balanced-budget allocation, for $\tau \in (0,1)$, derive the Marshallian and Hicksian elasticities of individual hours with respect to the after-tax wage $w_n = 1-\tau$, the balanced-budget elasticity $e_{BB} \equiv d\ln h / d\ln(1-\tau)$, and the Frisch elasticity. Order the four for every $\tau \in (0,1)$ and say which one Prescott's experiment uses.

Step — Marshallian. From (a) with w_n free, $h = \frac{1}{1+\alpha} - \frac{\alpha T}{(1+\alpha)w_n}$: $h_w \equiv \frac{\partial h}{\partial w_n} = \frac{\alpha T}{(1+\alpha)w_n^2}$, $h_T \equiv \frac{\partial h}{\partial T} = -\frac{\alpha}{(1+\alpha)w_n}$. At $T = \tau h$, $w_n = 1-\tau$: $e_M = \frac{w_n h_w}{h} = \frac{\alpha\tau}{(1+\alpha)(1-\tau)}$.

Step — Hicksian. Slutsky (the compensating transfer moves by $-h\,dw_n$): $\frac{\partial h^H}{\partial w_n} = h_w - h\,h_T = \frac{\alpha(T+w_n h)}{(1+\alpha)w_n^2} = \frac{\alpha c}{(1+\alpha)w_n^2}$; with $c = h$, $e_H = \frac{\alpha}{(1+\alpha)(1-\tau)}$.

Step — balanced budget and Frisch. $\ln h(\tau) = \ln(1-\tau) - \ln(1+\alpha-\tau)$ gives $e_{BB} = 1-h = \frac{\alpha}{1+\alpha-\tau}$. With the marginal utility of wealth λ fixed, $\frac{\alpha}{1-h} = \lambda w_n$ gives $\frac{dh}{d\ln w_n} = 1-h$: $e_F = \frac{1-h}{h} = \frac{\alpha}{1-\tau}$.

Step — ordering. $e_F = (1+\alpha)e_H > e_H > \tau e_H = e_M$; $e_H - e_{BB} = \frac{\alpha^2\tau}{(1+\alpha)(1-\tau)(1+\alpha-\tau)} > 0$; and $e_{BB} - e_M = \frac{\alpha\,N(\tau)}{(1+\alpha)(1-\tau)(1+\alpha-\tau)}$, with N from (b), so $e_{BB} \gtrless e_M$ iff $\tau \lesseqgtr \tau^*$. Why: $h'(\tau) = -h_w + h_T R'(\tau)$, so $e_{BB} = e_M + \frac{1-\tau}{h}\,|h_T|\,R'(\tau)$: below the peak a tax cut also cuts the transfer, an income effect that adds to the wage effect. So $e_F > e_H > e_{BB} > e_M$ for $\tau < \tau^*$; above τ^*, $e_{BB} < e_M$.

Step — which one. Prescott compares economies whose revenue is rebated: his experiment uses e_{BB}, and $R'(\tau^*) = 0$ reads $\tau^* = 1/[1+e_{BB}(\tau^*)]$. Kurlat's Exercise 7.5(j)–(l) computes e_F to compare it with micro estimates.

Economic intuition. They hold different things fixed: the transfer (e_M), utility (e_H), the budget (e_{BB}), the marginal utility of wealth (e_F); only e_{BB} lets the transfer move, so only its rank depends on the side of the peak.

Ref: Kurlat (2020), §7.3 and Exercise 7.5(j)–(l).

Grading rubric.

- Each elasticity: 0.5 pt (2 pts). Ordering, threshold τ^* proved, Prescott's choice: 1 pt.
- Cap: elasticities at a single τ (special case for the general one), max. half.

e) (3 points) *Parameter analysis.* Sign the effects of α and of τ on h, R, τ^*, R^*, x and e_{BB}, prove each sign and collect them in a table. Find the limits as $\alpha \to 0$ and $\alpha \to \infty$ and a bound on τ^* valid for every α, and read the results.

Step — direct signs. $\frac{\partial h}{\partial \alpha} = -\frac{1-\tau}{(1+\alpha-\tau)^2} < 0$, $h'(\tau) < 0$ (a); $\frac{\partial R}{\partial \alpha} < 0$, R' has the sign of $N(\tau)$ (b); $\frac{\partial e_{BB}}{\partial \alpha} = \frac{1-\tau}{(1+\alpha-\tau)^2} > 0$, $\frac{\partial e_{BB}}{\partial \tau} = \frac{\alpha}{(1+\alpha-\tau)^2} > 0$; $\frac{\partial x}{\partial \tau} > 0$ (c).

Step — Laffer peak. $\frac{d\tau^*}{d\alpha} = 1 - \frac{1+2\alpha}{2\sqrt{\alpha(1+\alpha)}} < 0$ because $(1+2\alpha)^2 = 4\alpha(1+\alpha)+1$; $R^* = (\sqrt{1+\alpha}+\sqrt{\alpha})^{-2}$ falls with α.

Step — welfare cost. With $a \equiv 1+\alpha$, $\ln(1-x) = \ln(1-\tau) + a\ln\frac{a}{a-\tau}$ and $\frac{\partial}{\partial a}\left[a\ln\frac{a}{a-\tau}\right] = \ln(1+z) - z < 0$ for $z \equiv \frac{\tau}{a-\tau} > 0$ ($\ln(1+z)-z$ is 0 at $z=0$ with derivative $-\frac{z}{1+z}$): x rises with α.

	$\partial/\partial\alpha$	$\partial/\partial\tau$
hours $h(\tau)$	$-$	$-$
revenue $R(\tau)$	$-$	$+$ iff $\tau < \tau^*$
Laffer peak τ^*, largest program R^*	$-$	0
welfare cost $x(\tau)$	$+$	$+$
elasticity e_{BB}	$+$	$+$

Step — limits and bound. $\alpha \to 0$: $\tau^* \to 1$, $R^* \to 1$, $h \to 1$, $x \to 0$, $e_{BB} \to 0$. $\alpha \to \infty$: $\sqrt{\alpha^2+\alpha} = \alpha + \frac{1}{2} - \frac{1}{8\alpha} + O(\alpha^{-2})$ gives $\tau^* \downarrow \frac{1}{2}$; $R^* \sim \frac{1}{4\alpha}$, $h \to 0$, $e_{BB} \to 1$, and $a \ln \frac{a}{a-\tau} \to \tau$ gives $x \to 1-(1-\tau)e^{\tau}$, a bound on the cost for every α. As $(\alpha+\frac{1}{2})^2 - \alpha(1+\alpha) = \frac{1}{4}$, $\tau^* \in (\frac{1}{2}, 1)$ for every α. Knife-edges: $\tau = 0$ (no first-order cost, $e_{BB} = e_H$); τ^* ($e_{BB} = e_M$).

Step — reading. A larger α makes hours more responsive: the base shrinks faster, the peak comes earlier (never below one half), the largest program shrinks and each rate costs more welfare; as $\alpha \to 0$ the tax is lump-sum. Prescott's account of Europe's lower hours needs a large α, an e_{BB} at the top of the micro estimates (Kurlat §7.3).

Ref: Kurlat (2020), §7.3 and Exercise 7.5.

Grading rubric.

- Signs, including $d\tau^*/d\alpha$ and $\partial x/\partial\alpha$, proved: 1.5. Table, limits, reading: 0.5 each.
- Caps: a sign without proof, 0 for it; analysis without proofs or reading, max. half.

Block 3 — Integrative: consumption, labor, pensions and welfare

(Kurlat chs. 2, 4, 6–7; Romer chs. 2, 8, 10 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) *Consumption × fiscal policy × credit.* Brazil's 2020 emergency aid, stylized. Households live two periods with utility $u(c_1) + \beta u(c_2)$ (u strictly concave, $\beta \in (0,1)$) and save a at the real rate $r > -1$ at which the government borrows, subject to a fixed limit $a \geq -b$, $b \geq 0$. A share $\lambda \in (0,1)$ of households is at the limit; the other $1-\lambda$ are strictly inside it. Every household gets a lump-sum transfer $T > 0$ in period 1, financed by debt repaid with interest through equal lump-sum taxes in period 2; the limit still binds for the constrained after the transfer. At the given r, average period-1 consumption per household changes by:

A. $\Delta\bar{c}_1 = T$

B. $\Delta\bar{c}_1 = (1-\lambda)T$

C. $\Delta\bar{c}_1 = 0$

D. $\Delta\bar{c}_1 = \lambda T$

Answer: D

Step — unconstrained. The debt T is repaid with taxes $(1+r)T$ in period 2. For endowments (y_1, y_2) the present value of resources is $y_1 + T + \frac{y_2-(1+r)T}{1+r} = y_1 + \frac{y_2}{1+r}$: unchanged. An unconstrained household keeps its plan and saves the whole transfer; its a rises by T, so the limit stays slack: $\Delta c_1 = 0$.

Step — constrained. At $a = -b$, $c_1 = y_1 + T + b$, so $\Delta c_1 = T$. The limit binds while $u'(y_1+T+b) > \beta(1+r)\,u'\big(y_2-(1+r)(T+b)\big)$; the left side falls and the right side rises with T, so it binds for every T below a threshold, as assumed.

Step — aggregate. $\Delta\bar{c}_1 = \lambda T + (1-\lambda)\cdot 0 = \lambda T$: the aggregate propensity to spend a debt-financed transfer is the constrained share. A treats every household as constrained; C applies Ricardian equivalence to all (it needs everyone to borrow at r); B makes both mistakes at once, giving the constrained Ricardian behavior and the unconstrained full spending.

Ref: Kurlat (2020), §6.2 (Taxes and Ricardian Equivalence) and Exercise 6.5; Romer (2012), §8.6 (Liquidity Constraints).

b) (3 points) *Labor × welfare measurement.* Jones and Klenow's flow utility (Kurlat §2.2, with $\sigma = 1$) is $\ln c - \theta(1-l)^2$, $\theta > 0$, where l is leisure and $1 - l = h$ is market hours. Between two dates, consumption per person in Brazil moves from c_0 to c_1 and market hours from h_0 to h_1 (both in $(0, 1)$), with survival and inequality unchanged. The consumption equivalent Λ, the factor on date-0 consumption (hours held at h_0) that delivers date-1 welfare, satisfies:

A. $\ln \Lambda = \ln(c_1/c_0)$

B. $\ln \Lambda = \ln(c_1/c_0) + \theta(h_0^2 - h_1^2)$

C. $\ln \Lambda = \ln(c_1/c_0) + \theta(h_0 - h_1)$

D. $\ln \Lambda = \ln(c_1/c_0) - \theta(h_0 - h_1)$

Answer: B

Step — definition. Kurlat's (2.2.2) with $u = \bar{u} + \ln c - \theta h^2$ ($\bar{u}$ cancels: survival is unchanged): $u(\Lambda c_0, h_0) = u(c_1, h_1)$, i.e. $\ln \Lambda + \ln c_0 - \theta h_0^2 = \ln c_1 - \theta h_1^2$, so $\ln \Lambda = \ln(c_1/c_0) + \theta(h_0^2 - h_1^2)$.

Step — reading. $\Lambda < c_1/c_0$ iff $h_1 > h_0$: growth that comes with longer hours overstates welfare growth. A ignores leisure; C uses hours in levels instead of squares; D makes two mistakes: levels, and the sign of the leisure term flipped, as if hours were a good.

Step — general case. The additive form needs $\sigma = 1$. With $\frac{c^{1-\sigma}}{1-\sigma}$, $\Lambda^{1-\sigma} = (c_1/c_0)^{1-\sigma} + (1-\sigma)\theta(h_0^2 - h_1^2)\, c_0^{\sigma-1}$ (when positive): the leisure term depends on the level c_0, and $\ln \Lambda$ tends to B as $\sigma \to 1$.

Ref: Kurlat (2020), §2.2 (Beyond GDP), eqs. (2.2.1)–(2.2.2).

c) (2 points, T or F) *Consumption × labor.* A household maximizes $\sum_{t=0}^{N} \beta^t [\ln c_t - v(h_t)]$ ($N \geq 1$, v increasing and strictly convex, interior hours) and borrows and lends freely at a given rate r. With these separable preferences, news at $t = 0$ of a permanent wage increase from $t = 1$ on, with w_0 unchanged, makes the household work more at $t = 0$.

Answer: F

Step — FOCs. With multiplier η on $\sum_t \frac{c_t - w_t h_t}{(1+r)^t} = a_0$: $c_t = \frac{[\beta(1+r)]^t}{\eta}$ and $v'(h_t) = \frac{\eta\, w_t}{[\beta(1+r)]^t}$, so h_t rises with η and with w_t.

Step — wealth effect. η solves $\Phi(\eta, w) \equiv a_0 + \sum_t \frac{w_t h_t - c_t}{(1+r)^t} = 0$. Φ is strictly increasing in η ($h_t \uparrow$, $c_t \downarrow$) and in each w_t ($w_t h_t \uparrow$), so the root is unique and $\frac{d\eta}{dw_t} = -\frac{\Phi_{w_t}}{\Phi_\eta} < 0$ for every $t \geq 1$.

Step — today. $v'(h_0) = \eta w_0$ with w_0 fixed: h_0 falls and $c_0 = 1/\eta$ rises. As in Kurlat's (7.4.4), hours today depend on $w_0 u'(c_0)$, and only $u'(c_0)$ moved: the household is richer and buys more leisure now, while intertemporal substitution shifts work toward the high-wage dates.

Ref: Kurlat (2020), §7.4 (A Dynamic Model), eqs. (7.4.2)–(7.4.4).

d) (2 points, T or F) *Search × social insurance.* In the steady-state search model (tightness θ, vacancy-filling rate $q(\theta)$ and job-finding rate $f(\theta) = \theta q(\theta)$ with $q' < 0 < f'$, productivity y, unemployment benefit $z < y$, flow vacancy cost κ, separation rate s, interest rate r, free entry of vacancies, Nash-bargained wage with workers' bargaining power $\phi \in [0, 1)$), more generous unemployment benefits (a higher z, e.g. a more generous *seguro-desemprego*) lower job creation θ even when workers have no bargaining power ($\phi = 0$).

Answer: T

Step — wage. With $\phi = 0$ the worker gets none of the match surplus: $V_E = V_U$. The asset equations $rV_E = w + s(V_U - V_E)$ and $rV_U = z + f(\theta)(V_E - V_U)$ then read $rV_E = w$ and $rV_U = z$, so $w = z$: the wage is the worker's outside option.

Step — free entry. $rV_F = y - w - sV_F$ and $0 = rV_V = -\kappa + q(\theta)V_F$ give $\frac{(y-z)\, q(\theta)}{r+s} = \kappa$. With $q' < 0$, $\frac{d\theta}{dz} = \frac{q(\theta)}{(y-z)\, q'(\theta)} < 0$; with Question 3(b)'s $q = A_m \theta^{-\xi}$, $\theta = \left[\frac{A_m (y-z)}{\kappa(r+s)}\right]^{1/\xi}$ and $\frac{\partial \ln \theta}{\partial z} = -\frac{1}{\xi(y-z)}$.

Step — why. Zero bargaining power does not shield the wage from z: the threat point moves it one for one, and the firm's surplus $y - z$ shrinks. For any $\phi \in [0, 1)$, job creation $(1-\phi)(y-z) = \kappa\left[\frac{r+s}{q(\theta)} + \phi\theta\right]$ has a right side increasing in θ, so again $d\theta/dz < 0$, and unemployment $s/[s + f(\theta)]$ rises.

Ref: Romer (2012), §10.6 (Search and Matching Models); Kurlat (2020), §7.5 (Search), eq. (7.5.1) with zero worker bargaining power.

Question 6. (15 points) Discursive. *Integrative: life-cycle saving × labor supply × pension design × the interest–growth gap.* Brazil's general regime (INSS) is pay-as-you-go (PAYG), with benefits computed from each worker's contribution record, while the minimum-wage floor and the non-contributory BPC pay amounts unrelated to it; the 2019 reform proposal included funded individual accounts (*capitalização*), dropped in Congress. A household lives two periods: in period 1 it works $h \in (0,1)$ at the wage $w > 0$, in period 2 it is retired. Utility is $u = \ln c_1 + \gamma \ln(1-h) + \beta \ln c_2$, with $\beta \in (0,1)$ and $\gamma > 0$. A payroll tax $\tau \in (0,1)$ finances a pension P; the budgets are $c_1 + a = (1-\tau)wh$ and $c_2 = (1+r)a + P$, with a free in sign. Each generation is $1+n$ times the previous one, wages are constant, $n, r > -1$, prices (r, w) are taken as given in (a), (b) and (d), and $r > n$ unless stated; write $q \equiv \frac{1+n}{1+r}$. Three designs: *funded*, $P = (1+r)\tau wh$; *Bismarckian PAYG*, $P = (1+n)\tau wh$, linked to the household's own contributions (it internalizes the link); *Beveridgean PAYG*, a lump sum P that the household takes as given, equal to $(1+n)\tau wh$ in equilibrium.

a) (3 points) *Funded and Bismarckian designs.* Write each lifetime budget as $c_1 + \frac{c_2}{1+r} = \omega wh$ and find ω. Solve the household problem (both consumptions, and hours h_F and h_{Bis}) and compare the two designs.

Step — lifetime budget (period budgets). Substitute $a = (1-\tau)wh - c_1$ into $c_2 = (1+r)a + P$ and divide by $1+r$: $c_1 + \frac{c_2}{1+r} = (1-\tau)wh + \frac{P}{1+r}$. Funded: $\frac{P}{1+r} = \tau wh$, so $\omega_F = 1$. Bismarck: $\frac{P}{1+r} = q\tau wh$, so

$$\omega = 1 - \tau + \tau q = 1 - \frac{\tau(r-n)}{1+r},$$

a weighted average of 1 and q with weight τ on q: $\omega \in (q, 1)$ when $r > n$, and $\omega > 0$ always. In both designs P moves with h, so ωw is what one more hour adds to lifetime resources.

Step — FOCs (lifetime budget). Add $\omega w(1-h)$ to both sides: $c_1 + \frac{c_2}{1+r} + \omega w(1-h) = \omega w \equiv X$, full income. The Lagrangian $u + \mu\big[X - c_1 - \frac{c_2}{1+r} - \omega w(1-h)\big]$ gives $\frac{1}{c_1} = \mu$, $\frac{\beta}{c_2} = \frac{\mu}{1+r}$, $\frac{\gamma}{1-h} = \mu\,\omega w$, where μ is the marginal utility of lifetime wealth; u is strictly concave and the budget linear, so the FOCs give the unique optimum. Spending on each good is its weight over μ; summing, $X = (1+\beta+\gamma)/\mu$.

Step — solution (FOCs and $X = \omega w$).

$$c_1 = \frac{\omega w}{1+\beta+\gamma}, \qquad c_2 = \frac{\beta(1+r)\,\omega w}{1+\beta+\gamma}, \qquad h = \frac{1+\beta}{1+\beta+\gamma}.$$

Hours depend on neither ω nor w: full income and the price of leisure are both proportional to ωw, so with log utility the income and substitution effects of ω cancel.

Step — comparison. $h_{Bis} = h_F$ and $c_t^{Bis} = \omega\, c_t^F$ for $t = 1, 2$. A Bismarckian system is a funded one whose contributions earn the growth rate n instead of r: with $r > n$ it is a pure wealth loss of $1 - \omega$ of lifetime income, with no distortion.

Economic intuition. The link between contributions and benefits keeps the leisure margin intact: the household treats the tax as saving at return n, so only its lifetime wealth changes.

Ref: Kurlat (2020), §6.2 (two-period consumption) and §7.2, §7.4 (labor–leisure choice, jointly with saving); Romer (2012), Problem 2.17 (social security in the Diamond model).

Grading rubric.

- Lifetime budget and ω, with $\omega \in (q, 1)$: 1 pt.
- FOCs, full income and the three closed forms: 1.5 pts. Hours quoted from the static model instead of derived: max. a third of the item.
- Independence of hours from ω and the scaling by ω: 0.5 pt.

b) (3 points) *Beveridgean design.* Find equilibrium hours h_{Bev}, compare them with h_F, explain the mechanism, and draw hours against $\tau \in (0,1)$ for the three designs.

Step — household problem (P given). The lifetime budget is $c_1 + \frac{c_2}{1+r} = (1-\tau)wh + \frac{P}{1+r}$: the price of leisure is now $(1-\tau)w$ and full income $X = (1-\tau)w + \frac{P}{1+r}$. The FOCs of (a), with ωw replaced by $(1-\tau)w$, give $(1-\tau)w(1-h) = \frac{\gamma X}{1+\beta+\gamma}$.

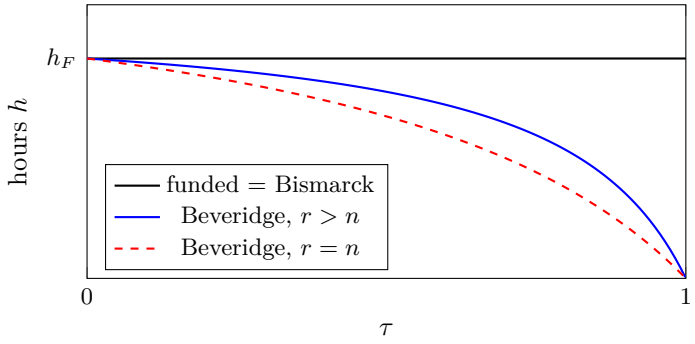
Step — equilibrium (pension budget, after the FOC). Impose $\frac{P}{1+r} = q\tau wh$, so $X = (1-\tau)w + q\tau wh$:

$$(1+\beta+\gamma)(1-\tau)(1-h) = \gamma(1-\tau) + \gamma q\tau h \implies h_{Bev} = \frac{(1+\beta)(1-\tau)}{(1+\beta)(1-\tau)+\gamma\omega},$$

using $1-\tau+q\tau = \omega$.

Step — comparison. With $x \equiv \frac{1-\tau}{\omega}$, $h_{Bev} = \frac{(1+\beta)x}{(1+\beta)x+\gamma}$, strictly increasing in x, and h_F is its value at $x=1$. Since $\omega - (1-\tau) = q\tau > 0$, $x < 1$ and $h_{Bev} < h_F$ for every r and n, including $r \le n$. As $\tau \to 1$, $h_{Bev} \to 0$.

Step — the wedge. In equilibrium an hour adds ωw to lifetime resources (budget above), but the household prices leisure at $(1-\tau)w$: it sees the tax and not the return $q\tau w$ that comes back as a pension. Relative to Bismarck, Beveridge taxes the marginal hour at the rate $q\tau/\omega$ and rebates the revenue lump sum (Prescott's tax–transfer experiment): the rebate offsets the income effect, and the substitution effect lowers hours.



Economic intuition. Beveridge breaks the link between the marginal contribution and the benefit, so the payroll tax becomes a labor tax; the curve for $r = n$ lies lower because a pension worth more in present value returns a larger share of each contribution as a lump sum that the household does not link to its hours (the wedge $q\tau/\omega$ rises with q).

Ref: Kurlat (2020), §7.2–7.3 (taxes and social security as Prescott's explanation of hours) and Exercise 7.5.

Grading rubric.

- FOC with P taken as given: 0.5 pt.
- h_{Bev} with the pension substituted after the FOC: 1 pt. Substituting before the FOC (the Bismarck problem): max. half of the item.
- Proof of $h_{Bev} < h_F$ and the wedge: 0.5 pt.
- Graph (scored separately): axes and labels; flat line at h_F for funded and Bismarck; Beveridge curve falling from h_F at $\tau = 0$ towards 0 as $\tau \to 1$: 1 pt.

c) (3 points) *Saving and capital.* Find private saving per unit of wage income, $a/(wh)$, in both PAYG designs, and the national saving rates s_F, s_{Bis} and s_{Bev} (private saving plus pension assets, per unit of wage income); rank them. If these are the saving rates of a Cobb–Douglas Solow economy with capital share $\alpha \in (0,1)$, the same n, technology growth g and depreciation δ ($\delta + n + g > 0$), by what factor does each PAYG design change steady-state output per effective worker relative to the funded design?

Step — full income per unit of wage income. In both PAYG designs $c_1 = \frac{X}{1+\beta+\gamma}$ (FOCs). Bismarck: $X = \omega w$ and $wh = \frac{(1+\beta)w}{1+\beta+\gamma}$. Beveridge: the hours equation of (b) gives $\frac{1-\tau}{h_{Bev}} = (1-\tau)+\frac{\gamma\omega}{1+\beta}$, so $\frac{X}{wh} = \frac{1-\tau}{h_{Bev}} + q\tau = \omega(1+\frac{\gamma}{1+\beta})$. In both designs,

$$\frac{X}{wh} = \frac{\omega(1+\beta+\gamma)}{1+\beta} \implies \frac{c_1}{wh} = \frac{\omega}{1+\beta}, \qquad \frac{a}{wh} = (1-\tau) - \frac{\omega}{1+\beta}.$$

Step — national saving rates. PAYG holds no assets, so national saving is the private saving above, the same in both designs: $s_{Bis} = s_{Bev} = (1-\tau) - \frac{\omega}{1+\beta} = \frac{\beta-\tau(\beta+q)}{1+\beta} \equiv s_{PAYG}$. Funded: $c_1 = \frac{wh}{1+\beta}$, private saving is $\frac{a_F}{wh} = \frac{\beta}{1+\beta} - \tau$ (negative if $\tau > \frac{\beta}{1+\beta}$: the household borrows against its pension), and adding the fund's assets τ gives $s_F = \frac{\beta}{1+\beta}$, independent of τ.

Step — ranking. $s_F - s_{PAYG} = \frac{\tau(\beta+q)}{1+\beta} = \tau - \frac{1-\omega}{1+\beta} > 0$: PAYG removes the fund ($-\tau$), and the household, poorer by $1-\omega$, cuts c_1 by $\frac{1-\omega}{1+\beta}$ and saves that much more, a partial offset.

Step — Solow (Kurlat ch. 4). Per efficiency unit of labor, $\tilde{y} = \tilde{k}^{\alpha}$ and the steady state solves $s\,\tilde{y} = (\delta+n+g)\tilde{k}$, so $\tilde{y}^* = \left(\frac{s}{\delta+n+g}\right)^{\frac{\alpha}{1-\alpha}}$; measuring saving out of output, $(1-\alpha)s$, changes nothing below. The common terms cancel, and both PAYG designs give the same factor ($j = Bis, Bev$):

$$\frac{\tilde{y}_j^*}{\tilde{y}_F^*} = \Big(\frac{s_{PAYG}}{s_F}\Big)^{\frac{\alpha}{1-\alpha}} = \Big[1 - \tau\Big(1+\frac{q}{\beta}\Big)\Big]^{\frac{\alpha}{1-\alpha}} \in (0,1) \qquad \text{for } \tau < \frac{\beta}{\beta+q},$$

the condition for positive PAYG saving. Per worker, Beveridge loses a further factor h_{Bev}/h_F: output per worker is $A\,h\,\tilde{y}$ (A: labor-augmenting technology).

Economic intuition. A funded system is forced saving that the household undoes one for one, so it is neutral for capital; a PAYG system pays the old out of the young's contributions, which never become capital. Both PAYG designs save the same share because they face the same interest rate and the same equilibrium lifetime resources per hour, ωwh, which the Euler equation splits in the same way.

Ref: Kurlat (2020), §4.2 and §4.5 (steady state per efficiency unit of labor), §6.2 (Euler equation); Romer (2012), Problem 2.17.

Grading rubric.

- $X/(wh)$ in both designs and $a/(wh)$: 1 pt. Shown for one design only: 0.5 pt.
- s_F, $s_{Bis} = s_{Bev}$ and the ranking, proved: 1 pt.
- Solow ratio with the positivity condition: 1 pt. Ratio not linked to the saving rates derived here (answered inside one model): max. half of the item.

d) (3 points) *Welfare.* At given prices, measure each PAYG design by its consumption-equivalent loss λ relative to funded: the share of funded consumption in both periods, at funded hours, whose loss leaves the household as well off as under the PAYG design. Find λ_{Bis} and λ_{Bev} in closed form, rank the three designs when $r > n$ (the two PAYG designs by a revealed-preference argument), and determine what changes when $r = n$ and when $r < n$.

Step — definition. Scaling both consumptions by $1-\lambda$ changes utility by $(1+\beta)\ln(1-\lambda)$, so λ_j solves $U_F + (1+\beta)\ln(1-\lambda_j) = U_j$.

Step — Bismarck (from (a)). Same hours, consumptions scaled by ω: $U_{Bis} = U_F + (1+\beta)\ln\omega$, so $\lambda_{Bis} = 1-\omega = \frac{\tau(r-n)}{1+r}$.

Step — Beveridge (from (b), (c)). $c_1 = \frac{\omega wh}{1+\beta}$ in both PAYG designs and $\frac{wh}{1+\beta}$ when funded, and $c_2 = \beta(1+r)c_1$ in all three, so $c_t^{Bev}/c_t^F = \omega h_{Bev}/h_F$. With $\frac{h_{Bev}}{h_F} = \frac{(1+\beta+\gamma)x}{(1+\beta)x+\gamma}$, $\frac{1-h_{Bev}}{1-h_F} = \frac{1+\beta+\gamma}{(1+\beta)x+\gamma}$, $\omega x = 1-\tau$ and $\rho \equiv \frac{1+\beta+\gamma}{1+\beta} > 1$:

$$1 - \lambda_{Bev} = \frac{\omega h_{Bev}}{h_F}\Big(\frac{1-h_{Bev}}{1-h_F}\Big)^{\frac{\gamma}{1+\beta}} = (1-\tau)\Big[\frac{(1+\beta+\gamma)\,\omega}{(1+\beta)(1-\tau)+\gamma\omega}\Big]^{\rho}.$$

Step — ranking by revealed preference. In equilibrium the Beveridge allocation satisfies $c_1 + \frac{c_2}{1+r} = (1-\tau)wh_{Bev} + q\tau wh_{Bev} = \omega wh_{Bev}$, so it is affordable under Bismarck. The Bismarck choice is the unique maximizer of the strictly concave u on that convex set, and $h_{Bev} \neq h_{Bis}$ by (b), so $U_{Bev} < U_{Bis}$: $\lambda_{Bev} > \lambda_{Bis}$. With $r > n$, $\omega < 1$ and $U_F > U_{Bis} > U_{Bev}$.

Step — $r = n$. $q = \omega = 1$: $\lambda_{Bis} = 0$, Bismarck replicates funded. Beveridge still distorts hours ($x = 1 - \tau < 1$), and by the previous step $\lambda_{Bev} = 1 - (1-\tau)\left[\frac{1+\beta+\gamma}{(1+\beta)(1-\tau)+\gamma}\right]^{\rho} > 0$.

Step — $r < n$ (dynamic inefficiency). PAYG now pays $n > r$: $\lambda_{Bis} = -\frac{\tau(n-r)}{1+r} < 0$, Bismarck beats funded. For Beveridge, with $b \equiv 1 - \frac{\gamma q}{1+\beta+\gamma}$ (so that $(1+\beta)(1-\tau) + \gamma\omega = (1+\beta+\gamma)(1-b\tau)$), let $L(\tau) \equiv \ln(1-\lambda_{Bev}) = \ln(1-\tau) + \rho \ln\omega - \rho\ln(1-b\tau)$. Then $L(0) = 0$ and

$$L'(\tau) = \frac{q}{\omega(1-b\tau)} - \frac{1}{1-\tau} = \frac{N(\tau)}{(1-\tau)\,\omega\,(1-b\tau)}, \qquad N(\tau) \equiv q(1-\tau) - \omega(1-b\tau).$$

N has degree at most 2, $N(0) = q - 1 > 0$ and $N(1) = -\frac{\gamma q^2}{1+\beta+\gamma} < 0$, so it has exactly one root $\tau_p \in (0,1)$: L rises on $(0,\tau_p)$ and falls on $(\tau_p, 1)$, with $L \to -\infty$ as $\tau \to 1$. Hence there is a unique $\bar\tau \in (\tau_p, 1)$, with no closed form, such that Beveridge beats funded if and only if $\tau < \bar\tau$.

Economic intuition. Bismarck costs only the return gap $1 - \omega$, which changes sign with $r - n$. Beveridge adds an hours distortion that survives at $r = n$; with $r < n$ a small Beveridge system still gains, because its return gain is first order in τ ($L'(0) = q - 1$) while the distortion is second order (see (e)).

Ref: Romer (2012), §2.11 (dynamic inefficiency) and Problem 2.17; Kurlat (2020), §4.3 (golden rule).

Grading rubric.

- Definition of λ and λ_{Bis}: 0.5 pt.
- λ_{Bev} in closed form: 1 pt. Quoted instead of derived: max. a third of the item.
- Revealed-preference ranking (affordability, uniqueness, $h_{Bev} \neq h_{Bis}$): 0.5 pt. Ranking asserted or read off numbers: 0 for that line.
- $r = n$, and $r < n$ with $\bar\tau$ proved to exist and be unique: 1 pt. "PAYG beats funded" claimed for Beveridge without the condition: 0 for that line.

e) (3 points) *Parameter analysis.* Sign the effects of r, n and τ on ω, h_{Bev}, s_{Bis}, s_{Bev}, the two Solow output ratios of (c), λ_{Bis} and λ_{Bev}; prove each sign and collect the signs in a table; find the limits $\tau \to 0$ (with the order in τ of $\lambda_{Bev} - \lambda_{Bis}$) and $r \to n$, and read the results.

Step — ω and λ_{Bis}. From $\omega = 1 - \tau + \tau q$: $\frac{\partial\omega}{\partial r} = -\frac{\tau(1+n)}{(1+r)^2} < 0$, $\frac{\partial\omega}{\partial n} = \frac{\tau}{1+r} > 0$, $\frac{\partial\omega}{\partial\tau} = q - 1 = -\frac{r-n}{1+r}$, with the sign of $n - r$. $\lambda_{Bis} = 1 - \omega$ has the opposite signs.

Step — h_{Bev}. $\frac{\partial h_{Bev}}{\partial x} = \frac{(1+\beta)\gamma}{[(1+\beta)x+\gamma]^2} > 0$, and from $x = \frac{1-\tau}{\omega}$: $\frac{dx}{d\tau} = \frac{-\omega-(1-\tau)(q-1)}{\omega^2} = -\frac{q}{\omega^2} < 0$, $\frac{\partial x}{\partial r} = -\frac{x}{\omega}\frac{\partial\omega}{\partial r} > 0$, $\frac{\partial x}{\partial n} = -\frac{x}{\omega}\frac{\partial\omega}{\partial n} < 0$. So h_{Bev} falls with τ for any (r,n), rises with r and falls with n.

Step — saving and output. From $s_{Bis} = s_{Bev} = s_{PAYG} = \frac{\beta - \tau(\beta+q)}{1+\beta}$ (c): $\frac{\partial s_{PAYG}}{\partial r} = \frac{\tau(1+n)}{(1+\beta)(1+r)^2} > 0$, $\frac{\partial s_{PAYG}}{\partial n} = -\frac{\tau}{(1+\beta)(1+r)} < 0$, $\frac{\partial s_{PAYG}}{\partial\tau} = -\frac{\beta+q}{1+\beta} < 0$. Since s_F is constant and $z \mapsto z^{\alpha/(1-\alpha)}$ is increasing, both output ratios have the same signs.

Step — λ_{Bev} in r and n. At fixed τ, r and n enter $1 - \lambda_{Bev}$ only through ω, and

$$\frac{\partial \ln(1-\lambda_{Bev})}{\partial\omega} = \frac{\rho}{\omega} - \frac{\rho\gamma}{(1+\beta)(1-\tau)+\gamma\omega} = \frac{\rho(1+\beta)(1-\tau)}{\omega\,[(1+\beta)(1-\tau)+\gamma\omega]} > 0,$$

so λ_{Bev} rises with r and falls with n, like λ_{Bis}.

Step — λ_{Bev} in τ. Write $1 - \lambda_{Bev} = \omega\, e^{\varphi(x)}$ with $\varphi(x) \equiv \ln x + \rho \ln\frac{1+\beta+\gamma}{(1+\beta)x+\gamma}$: $\varphi(1) = 0$ and $\varphi'(x) = \frac{\gamma(1-x)}{x[(1+\beta)x+\gamma]} > 0$ on $(0,1)$, so $e^{\varphi(x)} = \frac{1-\lambda_{Bev}}{1-\lambda_{Bis}} < 1$, the analytic twin of (d). Then $L'(\tau) = \frac{q-1}{\omega} - \varphi'(x)\frac{q}{\omega^2} < 0$ if $r \geq n$: λ_{Bev} rises with τ. If $r < n$, λ_{Bev} falls on $(0,\tau_p)$ and rises on $(\tau_p, 1)$, by (d).

	r	n	τ
ω	$-$	$+$	sign of $n-r$
h_{Bev}	$+$	$-$	$-$
$s_{Bis}=s_{Bev}$, both output ratios	$+$	$-$	$-$
λ_{Bis}	$+$	$-$	sign of $r-n$
λ_{Bev}	$+$	$-$	$+$ if $r\geq n$; $-$ then $+$ (at τ_p) if $r<n$
$h_F=h_{Bis}$, s_F	0	0	0

Step — limit $\tau\to 0$. $\omega, x\to 1$: $h_{Bev}\to h_F$, $s_{PAYG}\to s_F$, both output ratios $\to 1$, and $\lambda_{Bis},\lambda_{Bev}\to 0$. $\lambda_{Bis}=(1-q)\tau$ is first order, while $\lambda_{Bev}-\lambda_{Bis}=\omega\big(1-e^{\varphi(x)}\big)$, with $\varphi(1)=\varphi'(1)=0$, $\varphi''(1)=-\frac{\gamma}{1+\beta+\gamma}$ and $x-1=-\frac{q\tau}{\omega}$, gives

$$\lambda_{Bev}-\lambda_{Bis}=\frac{1}{2}\,\frac{\gamma}{1+\beta+\gamma}\,(q\tau)^2+O(\tau^3)\,:$$

a Harberger triangle, one half times the leisure share of full income times the squared wedge $q\tau$.

Step — limit $r\to n$. $q,\omega\to 1$: $\lambda_{Bis}\to 0$, but $h_{Bev}\to\frac{(1+\beta)(1-\tau)}{(1+\beta)(1-\tau)+\gamma}<h_F$, λ_{Bev} stays positive by (d), and $s_{PAYG}\to s_F-\tau$: at $r=n$ Bismarck is harmless to the household and still lowers saving by the whole contribution.

Economic intuition. The interest–growth gap prices the first-order cost of PAYG, $1-\omega=\tau(r-n)/(1+r)$; the benefit–contribution link decides the second-order cost in hours, which only Beveridge pays and whose leading term grows with γ and τ. A higher r makes PAYG costlier but, by lowering the present value of the pension, raises Beveridge hours and PAYG saving. Saving is lower under PAYG for every (r,n), so the capital cost does not vanish where the household's cost does.

Ref: Kurlat (2020), §4.3 and §7.3; Romer (2012), §2.11.

Grading rubric.

- Signs of ω, h_{Bev}, $s_{Bis}=s_{Bev}$, the output ratios and λ_{Bis}, each proved: 1 pt.
- Signs of λ_{Bev} in r, n, and in τ by case, proved: 0.5 pt.
- Table of signs: 0.5 pt. Limits, including the second-order term: 0.5 pt.
- Reading: 0.5 pt. Signs listed without proofs or without the reading: max. half of the item.